

Project Problem and Success Test

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Permissions and Human Review

Without opening the old lesson, explain one key idea from Permissions and Human Review. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Evidence and Sources

Without opening the old lesson, explain one key idea from Evidence and Sources. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Image and Voice Pipelines

Without opening the old lesson, explain one key idea from Image and Voice Pipelines. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Write an acceptance case for Planning the application. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Planning the application. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Plan independently

Plan a 15-minute fictional club chooser. Specify schema, score, duration limit, tie rule, invalid-data behavior, output and three acceptance cases.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Specify three acceptance tests for a recommender: a normal match, a tied score and no qualifying book. Include expected behaviour before implementation.

6. Solve the given case

A project recommends from ten approved books using genre and length.

Define acceptance tests.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Define a testable goal for a book recommender serving readers who want short space books. Include a concrete success condition.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 32 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Freeze the brief before building so later choices can be traced to requirements.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Specify three acceptance tests for a recommender: a normal match, a tied score and no qualifying book. Include expected behaviour before implementation.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Name the user need

State the purpose and decision owner of the worked chooser.

My reasoning and evidence

13. W2 Write an acceptance case

Give a tie case for A and B with score 8 each and valid durations.

My reasoning and evidence

14. W3 Repair vague scope

Replace the requirement always choose the best activity with a checkable one.

My reasoning and evidence

15. W4 Plan independently

Plan a 15-minute fictional club chooser. Specify schema, score, duration limit, tie rule, invalid-data behavior, output and three acceptance cases.

My reasoning and evidence

16. W5 State non-goals

Name two capabilities excluded from this prototype and explain why.

My reasoning and evidence

17. A reproducible data investigation — W1 State the question

Write the worked investigation question so its population of records is clear.

My reasoning and evidence

18. A reproducible data investigation — W2 Trace exclusions

Classify 0, None, -2 and 60 under the schema.

My reasoning and evidence

19. A reproducible data investigation — W3 Repair a denominator

A report divides a valid total of 24 by six supplied rows, though only three are valid. Correct the valid-record mean.

My reasoning and evidence

20. A reproducible data investigation — W4 Investigate independently

Analyze raw durations [6, 12, None, 0, 70, 18] with the same schema. Report question, accepted list, rejections, mean and one limitation.

My reasoning and evidence

21. A reproducible data investigation — W5 Support reproduction

A report gives only mean 9. What additional evidence lets a peer reproduce it?

My reasoning and evidence

22. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

23. Pictorial case practice: Project Problem and Success Test

A fictional reading-break chooser needs a transparent contract.

Items have id text, minutes integer 0–30, interest integer 0–5. Request limit is integer 0–30.

Eligible if minutes ≤ limit; score = interest; highest wins; ties choose alphabetically smaller ID; invalid rows excluded and counted.

Project Problem and Success Test

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Specify inputs

Validate each row and request; keep rejected-row reasons.



2. Filter and rank

At limit 8: A(time8,interest3), B(9,5), C(6,3). B is ineligible.



3. Expected output

A beats tied C by ID; report A and its score 3.



4. Acceptance cases

Also plan invalid request, no eligible item and equal-score cases.

CHANGE ONE THING

Reduce the limit from 8 to 5.

Explain the changed result using the matching lesson.

Write an equality test and explain which requirement it checks.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Retain the Transparent Book Recommender's schema, constraints, score, tie rule and acceptance table as its build specification.

24. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

Retain the Transparent Book Recommender's schema, constraints, score, tie rule and acceptance table as its build specification.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

25. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional reading-break chooser needs a transparent contract.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Also plan invalid request, no eligible item and equal-score cases.

1. Filter and rank
2. Specify inputs
3. Acceptance cases
4. Expected output

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Acceptance cases Also plan invalid request, no eligible item and equal-score cases. Planning fixes what success means before code returns an answer. The duration boundary is including equality, and the tie policy prevents accidental list-order dependence. A complete output may include why other rows were excluded. Test expectations should come from the contract rather than copying the first program result.

Practice 2: explain the deciding evidence

Changed-case practice: Reduce the limit from 8 to 5. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. Return no eligible item.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Return no eligible item. All three valid rows take more than five minutes.

Repair a missing building block

Review Permissions and Human Review (ai-2/31). Goal: Explain the mechanism and terminology of least privilege and tool scopes. Short practice: Which proposal is allowed, and why is an authorised object alone insufficient?. Return to this week and explain what you changed.

Review Evidence and Sources (ai-2/29). Goal: Explain the mechanism and terminology of claim support and citation evaluation. Short practice: Why is the original answer not fully supported despite having citations on every claim?. Return to this week and explain what you changed.

Review Image and Voice Pipelines (ai-2/26). Goal: Explain the mechanism and terminology of speech pipelines and multimodal evidence. Short practice: Which stage first fails, and why should the faulty transcript not be silently overwritten?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

